

Bedo Hydraulic Machines Vocational Training

Unity Demo Evaluation Summary

by Husni Hawash - October 30, 2025

This evaluation highlights significant shortcomings in the current Unity demo, primarily impacting user experience, interaction, and visual presentation.

Key Areas for Improvement:

1. User Experience (UX):

- a. **Outdated Design:** The overall look and feel of the UI and virtual space are outdated, failing to engage modern users.
- b. **Lack of Training Basics:** The demo doesn't adequately guide inexperienced users on how to interact with the lessons, creating a steep learning curve.

2. User Interaction:

- a. **Poor Navigation:** Camera controls are severely limited, making it difficult for users to properly view the machinery and experience the environment.
- b. **Limited Interactivity:** The demo relies solely on basic clicks, lacking essential features like drag-and-drop.
- c. **No Feedback:** Users receive zero visual or auditory feedback on their actions or clicks.
- d. **Missing Action Feedback:** Critical visual feedback, such as water flow after adjusting values, is absent.
- e. **Instability & Bugs:** The demo is highly buggy and unstable, leading to a frustrating experience.

3. User Interface (UI):

- a. **Inconsistent Design:** The UI suffers from inconsistent design direction, fonts, text sizes and style.
- b. **Outdated Appearance:** The UI appears visually outdated and lacks a modern aesthetic.
- c. **Flawed Modal & Notification Systems:** The modal view system is incorrectly implemented, and notifications are poorly positioned on the screen.
- d. **Missing Progress Indicators:** There is no visible timeline or progress indicator for steps.
- e. **Inefficient 3D Info Cards:** 3D info cards are excessively large yet contain minimal content.

4. Rendering and Visual Quality:

- a. **Unnatural Lighting:** The absence of baked lightmaps results in unnatural and unrealistic lighting.
- b. **Incorrect Environment Setup:** The environment sphere is improperly assigned, leading to visual errors (e.g., environment assigned to a glass window).
- c. **Lack of Realism in Materials:** Materials lack PBR (Physically Based Rendering) shading, preventing realistic surface representation.

In essence, the demo requires a comprehensive overhaul to become an effective and engaging training tool, addressing fundamental issues in usability, interaction design, and visual fidelity.

Execution timeline for the remake of the demo lesson (POC):

Phase 1: Planning & Foundation (Approx. 1 Week)

- Project Scoping & Prioritization
- UX Research & Mockups
- Technical Design & Architecture

Phase 2: Core UI, Interaction & Simulation Programming (4 Weeks)

- UI Programming & Integration
- Interaction Programming
- Simulation Programming (Animation)

Phase 3: Visual Fidelity & Polish (3 Weeks - in parallel)

- Rendering & Look Development
- Model, Texturing and shading enhancement
- Performance Optimization

Phase 4: Testing, Refinement & Deployment (1 week)

- Comprehensive QA Testing

Proposed Tech stack for the POC:

- Online web deployment only (desktop browser)
- React, Threejs (webGL), Redux (no backend)

Financial:

- Initial dry cost for the demo POC lesson will be between 4,500 to 6,000 JDs

Note:

- **Full platform study will require a cost**